



Introduction to Database Systems

Project Requirements

Project Description

Objectives

This project should make the student able to:

- ☐ Go through the phases of creating a relational-database-based system.
 - a. Analysis phase
 - b. Design phase
 - c. Implementation phase
- ☐ Use database management system (Microsoft SQL server)
- ☐ Create database web application
- ☐ Work in a team and learn how to communicate and organize work with others

Project Phases

<i>Project Phase</i>	<i>Deliverables</i>	<i>%</i>	<i>Deadline</i>
Analysis Phase	Project Proposal	5%	Week4 : 17th October 2025
Design Phase	ER Diagram	20%	- Week 5 (Initial Version – 8%): 24 October 2025 - Week 7 (Final version – 12%): 7th November 2025
	Database Schema	10%	Week 9: 21 th November 2025
Implementation Phase	Project Phase 1	15%	Week10 : 28 th November 2025
	Project Phase 2	50%	Week 13: 19 th December 2025
	Project Discussion		Week 14: 26 th December 2025 (During lab time)

Notes:

- ☐ The submission deadline will be **Friday 11:59 pm** for all phases in their respective weeks
- ☐ All project phases will be submitted to the classroom by the team leader and must be named as:
CSAI202_F25_T[TeamNumber]_[PhaseNumberName]

Project Requirements

It is required to create a relational-database-based web application. Your application should be based on a *real-world* model. You will go through the following phases.

1- Team Formation Phase

Number of students per team is 3 to 4 students.

Note: *Workload distribution should be vertical. A student is expected to work across all layers in the application.*

2- Analysis Phase

- ☐ Your team should select a *real-world* application that uses a database for its operation.
- ☐ Identify the requirements for your database. You may:
 - a. Meet people who use the real-world model of your system,
 - b. Get a list of requirements from a documented system, or
 - c. Identify reasonable and acceptable requirements by yourself.
- ☐ Applications are not supposed to be neither *too complex nor trivial*.
- ☐ Think of non-conventional applications.

Prepare and deliver a *project proposal* report.

3- Design Phase

In this phase, you will identify the structure of the database that will be used in your system (You will be a database designer) as well as Graphical User Interface pages and components (You will be a web designer). This phase includes 3 submissions:

1. ER Diagram (Initial Version)

- ☐ Identify the database entities and the relationships between these entities. This is done by thinking about the real-world model of your system.
- ☐ Draw the "Entity-Relationship" (ER) diagram for your database or the "Extended Entity-Relationship" (EER) if applicable.
- ☐ A good, clear and a well-defined ER diagram will make it easier for you when creating your tables and relationships.

Prepare and deliver the *ER diagram* report.

Your ER report will be evaluated by instructors to correct any design errors

2. ER Diagram (Final Version)

- ☐ Re-Draw the ER Diagram if needed (based on the instructors' feedbacks)

Prepare and deliver the Final *ER diagram* report.

3. Schema Design

- ☐ Starting from the corrected ER report
 - a. Follow the algorithm described in the textbook to map the diagram into database relations (schema)
 - b. Show primary and foreign keys (**Note:** *You can ask for TAs Feedback before creating the database*)
 - c. Create the database using a database engine (Microsoft SQL Server)

Prepare and deliver the *database schema* report.

4- Implementation Phase

In this phase you will convert your design into a working system. This phase implies the development of the ***user interface*** (You will be a front-end web developer) as well as the ***application logic*** and the integration between all layers (You will be a back-end web developer). This phase includes 2 main submissions:

1. Project Phase 1:

- ☐ Develop a friendly GUI for all the screens of your project.
- ☐ Fill the table of the database with random data.
- ☐ Create all necessary queries that you will need in your application as SQL file in the DBMS.

Deliver the screens of your application as a project and a copy of the database with the SQL queries.

2. Project Phase 2

- ☐ Develop application logic.
- ☐ Integrate between application layers.

Deliver the following items:

- **The complete project files.**
- **A backup of the database.**
- **A short video to explain a working demo of your application (5 mins max)**

Analysis Phase Deliverables

Project Proposal Report

Report should contain:

- 1- Cover page. (described below)
- 2- Proposed project description: one or two paragraphs to describe the project (the system) you intend to implement.
- 3- List of system users: who can use your system? (2-4 types of users)*
- 4- For each user, write functionalities that your system provides (5-10 functionalities per user type)*
- 5- List of real-world entities that should be present in your system. (10-15 entities)*

Notes:

- ☐ It is required to write **three** different project ideas ordered from the most idea you prefer.
- ☐ Storing data is **NOT** functionality; rather it is what users will do with stored data.
- ☐ The above numbers are for guidance and may be somehow tolerated.

Design Phase Deliverables

1- ER Diagram Report (Initial & Final)

Report should contain

- 1- Cover page. (described below)
- 2- Problem definition.
- 3- List of system users and privileges of each user.
- 4- List of entities and a brief description for each entity. This doesn't mean to list the attributes of the entity. It means to describe what this entity represents in the database.
- 5- List of relationships and a brief description for each relationship.
- 6- ER Diagram.
 - a. The first page should show the entities and the relationships between all the entities (with no attributes on entities. Relationships attributes should be shown here).
 - b. The rest pages should show the attributes of each entity.

Notes:

- ☐ *State explicitly any reasonable assumptions or restrictions you have.*
- ☐ *You must use a drawing tool to draw the ER Diagram (not in a paper)*

2- Database Schema Report

Report should contain:

- 1- Cover page. (described below)
- 2- The final approved ER diagram
- 3- Database schema diagram showing
 - a. Database relations (tables) showing primary keys.
 - b. Foreign keys showing the referenced relations. (Can be shown as arrows from referencing relation to referenced relation)

Implementation Phase Deliverables

1- Project Phase 1

Submission should contain:

- 1- Softcopy of fully developed GUI pages.
- 2- A backup of the database after adding random data in all tables- At least 30-40 tuples in major tables.
- 3- A .sql file with all necessary queries that you will need in your application.
- 4- Report with all application screens and the functionality of each screen.

2- Project Phase 2

Release should contain

- 1- Softcopy of the project (.zip file):
 - a. Logic fully developed + input data validation.
 - b. Layers fully integrated.
 - c. Database creation scripts with ready sample data.
- 2- Project demo video.
- 3- Report with following:
 - a. All delivered reports in one document divided as analysis, design, and implementation sections.
 - b. Workload distribution for the entire project.

Final Delivery Evaluation Criteria

☐ User support (15%)

- Different types of users supported by your application.
- No hardcoded passwords. Users' data should be stored in a DB table.
- Each individual user should have a username and a password.
- Application should enable users to sign up, login, logout and change password.
- Only the first admin user can be created manually then he should be able to change his password.
- Admin should be able to create other admins through the application.
- Encrypted password is recommended.

☐ System Functionality (50%)

- How comprehensive is the functionality supported by your application for different types of end users.
- All access to the database must be done through the application. Direct access to the database table is not allowed.

☐ Reporting Facility (20%)

- Statistical reports depending on your application.
 - Detailed statistical reports for specific parts of the database
 - Managerial level reports to see some overall statistics the whole application

☐ GUI (15%)

- Do not expect users to be database programmers.
- Do not expect users to memorize IDs. Use names instead of IDs.
- Interfaces should be operation-oriented not table oriented. i.e. categorize your end user interface items with respect to functionalities to be supported rather than tables to be accessed.

☐ Individual's role: Grade for each individual according to his/her role.

Important Note

- Each team member should identify EXACTLY his/her role in the project; as he/she will be evaluated accordingly, not necessarily that all team members got the same mark.

Project Deliverable Documents

Cover Page

All project deliverables should have the following cover page

Zewail City of Science, Technology and Innovation University of Science and Technology School of Computational Sciences and Artificial Intelligence CSAI 202 - Fall 2023	<Write here team number> example1: team 07
Introduction to Database Systems <Project Name> <Report Title (Deliverable Description)> Team Number: #	
Team Members: <Name>	<ID>

Contact info:

Write one email to be able to contact your team.

<Date>

Appendix A – Super entity and sub entity (Inheritance)

This appendix explains how to handle inheritance (super-sub entity) relationship during project design phase.

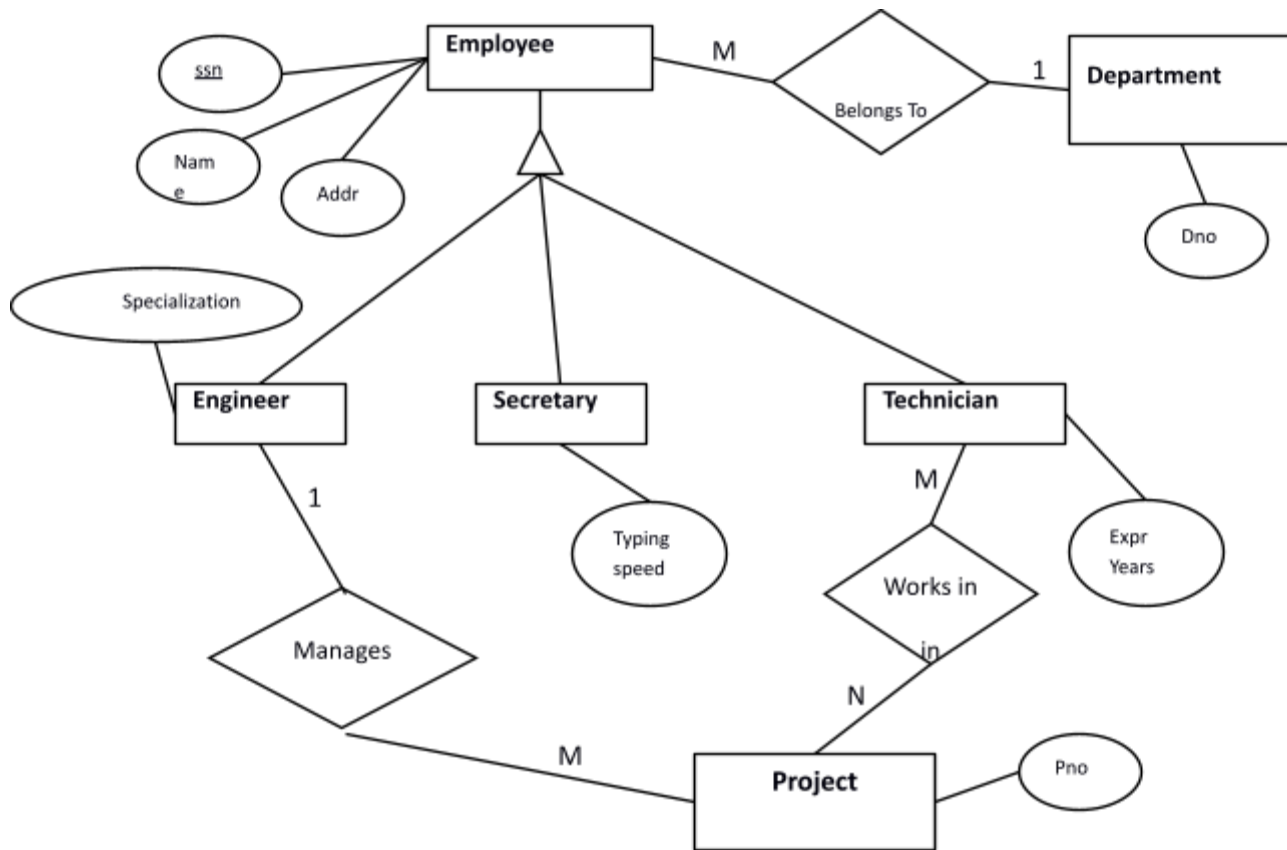
Example:

Assume you are designing a database for a company with the following partial requirements:

- ☐ ALL employees has ssn, name, Addr,.....etc. and belongs to a department
- ☐ An engineer is an employee with a specialization and manages company projects
- ☐ Other types of employees are secretary with a typing speed and technician who works in projects

Here we have three types of employees with some common attributes and relationships but each type also has its own different attributes and/or relationships

To represent such a case in ER diagram, make employee a super entity and derive three sub entities from it as shown below:



Here entity employee is a super entity as it has attributes and relationships common to Engineer, Secretary, and Technician entities. Each sub entity inherits both the attributes and relationships from its super entity. In addition, it has the same key (as will be shown in the schema below)

The super entity is connected to its sub entities through a triangle as shown in the figure. Please use the same notation in your project ER.

Mapping super entity and sub entity into schema

Super-sub entity relation may be considered as a special type of one-to-one relationship as each sub-entity is related to exactly one super

So the above ER mapping to a schema is:

Department

<u>Dno</u>	...rest of Dept table
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Project

<u>Pno</u>	...rest of Project table	Mngr_ssn
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Employee

<u>ssn</u>	Name	Addr	Dno	rest of Emp table
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Engineer

<u>ssn</u>	Specialization	rest of Eng table
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Secretary

<u>ssn</u>	Typing Speed	rest of Sec. table
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Technician

<u>ssn</u>	Experience Years	rest of Tech. table
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Works_IN

<u>Tssn</u>	<u>Pno</u>
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Foreign keys: (shown in bold)

- Employee (Dno) references Department (Dno)
- Project (Mngr_ssn) references Engineer (ssn)
- Works_IN (Tssn) references Technician (ssn)
- Works_IN (Pno) references Project (Pno)

- Engineer (ssn) references Employee (ssn)
- Secretary (ssn) references Employee (ssn)
- Technician (ssn) references Employee (ssn)

Note:

The last three keys are foreign keys from the super-table and at the same time the primary keys in the sub-tables. This is one possible representation of one-to-one relationships. Other possible representations are applicable too.